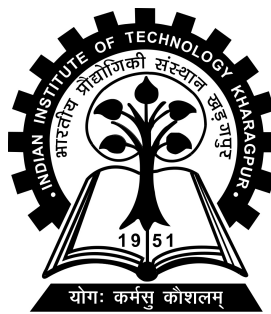


FarmFederate: Multimodal Federated Learning for Tea Leaf Disease Detection

Project-II report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Bachelor of Technology
in
Mechanical Engineering

by
Ayush Debnath
(22ME31010)

Under the supervision of
Professor Sudip Misra



Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

Spring Semester, 2025-26

April 27, 2026

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

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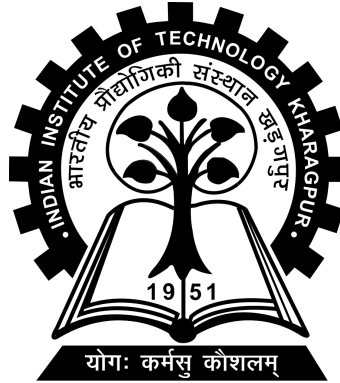
Place: Kharagpur

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**DEPARTMENT OF COMPUTER SCIENCE AND
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CERTIFICATE

This is to certify that the project report entitled “**FarmFederate: Multimodal Federated Learning for Tea Leaf Disease Detection**” submitted by **Ayush Debnath** (Roll No. 22ME31010) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Mechanical Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

	Professor Sudip Misra
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Abstract

Name of the student: **Ayush Debnath**

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Thesis title: **FarmFederate: Multimodal Federated Learning for Tea Leaf Disease Detection**

Thesis supervisor: **Professor Sudip Misra**

Month and year of thesis submission: **April 27, 2026**

Field-level diagnosis of tea leaf diseases traditionally relies on a combination of visual inspection and agronomist logs. However, automated diagnostic tools generally focus exclusively on either imagery or text, ignoring their combined potential. Furthermore, because agricultural data is often fragmented across multiple estates with limited connectivity, transferring raw data to centralised servers poses logistical and privacy challenges. To address this, we introduce FarmFederate, a multimodal federated learning framework designed to classify five common tea pathogens (*leaf blight*, *leaf hoppers*, *leaf rust*, *looper caterpillars*, *mosquito bug*). A key feature of our approach is a diagnostic fallback mechanism that addresses field data scarcity by mapping textual symptom descriptions to reference images annotated with Oriented Bounding Boxes (OBB), thereby improving visual explainability. The architecture connects lightweight Large Language Model (LLM) text encoders with Vision Transformer (ViT) image encoders across eight distinct Vision-Language Model (VLM)

fusion strategies. Distributed training is coordinated via Federated Averaging (FedAvg), ensuring that sensitive field records remain fully localised. Evaluated on a localised corpus with 200 field photographs, 371 YOLO-OBB annotations, 792 training-run image tensors after minority-class fill, and 3,000 agronomic text samples, our empirical results on a Tesla T4 GPU confirm the effectiveness of the multimodal approach. The best text-only and image-only baselines reach macro F1 scores of 0.487 and 0.911, while the best VLM fusion, CLIP-style contrastive fusion, reaches 0.949. Across three simulated non-IID clients, the federated training protocol preserves centralised performance while keeping raw field records local. Ultimately, FarmFederate ranks within the top tier of evaluated tea disease classifiers relative to a 35-model benchmark, bypassing the contemporary mean F1 threshold of 0.892.

Contents

Declaration	i
Certificate	ii
Abstract	iii
Contents	v
List of Figures	vii
List of Tables	ix
1 Introduction	1
1.1 Motivation	2
1.2 Contributions	3
2 Related Work	4
2.1 Tea Leaf Disease Detection	4
2.2 Federated Learning for Tea and Plant Disease Detection	5
2.3 Vision-Language Models for Agriculture	6
3 System Model	8
3.1 Architecture Overview	8
3.2 Dataset Description	9
3.2.1 Image Data	9
3.2.2 Text Data	10
3.3 Model Variants	11
3.3.1 LLM Text Encoders	11
3.3.2 ViT Image Encoders	11
3.4 Problem Formulation	12
3.5 Loss Function Formulation	13
4 FarmFederate Framework	16
4.1 VLM Fusion Architectures	16

4.2	Federated Learning with FedAvg	16
4.3	Anti-Collapse Mechanisms	18
5	RAG Diagnosis Module	20
5.1	Architecture	20
5.2	RAG Scoring Formulation	22
5.3	Text-to-Image Visual Fallback	23
5.4	Knowledge Base	24
5.5	RAG Evaluation	24
6	Performance Evaluation	26
6.1	Experimental Setup	26
6.2	Phase 1: Disease-Biased Dataset Comparison	27
6.3	Phase 2: Complete Model Comparison	29
6.3.1	LLM Encoder Results	29
6.3.2	ViT Encoder Results	30
6.3.3	VLM Fusion Results	30
6.3.4	Federated vs. Centralised Training	31
6.3.5	Per-Class Analysis	32
6.3.6	Modality Ablation	32
6.3.7	Statistical Significance	33
6.4	Discussion and Benchmark Placement	34
7	Conclusion	37

List of Figures

3.1	FarmFederate architecture: multimodal federated learning pipeline (top) and RAG diagnosis pipeline (bottom).	14
3.2	Compute allocation: tea garden edge devices (green) run leaf image encoding, text encoding, VLM fusion, disease classification, and per-estate FAISS retrieval locally; the global server (blue) aggregates model weights with FedAvg. Raw leaf images, agronomist annotations, and knowledge-base documents never leave the garden device.	15
3.3	Tea disease class distribution in the field-collected YOLO-OBB annotations.	15
3.4	Dataset: 200 field images, 371 OBB boxes, 792 training-run image tensors, and 3 000 text samples.	15
3.5	Parameter ranges across 18 architectures. VLM fusions heaviest; ViT lightest.	15
3.6	Dataset statistics and model sizes: disease-class distribution (left), field-data augmentation summary (centre), and parameter ranges (right).	15
4.1	Eight VLM fusion architectures implemented in FarmFederate.	17
5.1	RAG Diagnosis Module: query encoded by Sentence-BERT, top- k passages retrieved via FAISS, optionally re-ranked by VLM, and confidence adjusted by IoT readings.	21
5.2	FarmFederate visual diagnostic fallback. When a text-only prompt is provided, the system retrieves and annotates a representative image using YOLO OBB labels, providing immediate visual explainability.	24
5.3	Retrieved disease-type distribution.	25
5.4	IoT-adjusted confidence distribution.	25
5.5	Knowledge-base class balance.	25
5.6	Retrieval score spread by class.	25
5.7	Additional RAG diagnostics from the generated result bundle: retrieved evidence distribution, IoT confidence adjustment, knowledge-base balance, and score stability.	25
6.1	RAG scores.	30
6.2	LLM F1.	30

6.3	ViT F1.	30
6.4	VLM F1.	30
6.5	Retrieval and encoder comparison: RAG cosine scores, LLM F1, ViT F1, and VLM fusion F1.	30
6.6	RAG score heatmap.	31
6.7	Model performance heatmap across 18 architectures.	31
6.8	Heatmap comparison: RAG retrieval score matrix (left) and full model performance across LLM, ViT, and VLM groups (right).	31
6.9	Statistical significance analysis across LLM, ViT, and VLM model families.	33
6.10	Accuracy-per-parameter efficiency.	35
6.11	Model size versus F1.	35
6.12	Best model from each family.	35
6.13	Overall model ranking.	35
6.14	Additional analysis from the exported run: efficiency, size-performance trade-off, best-family comparison, and full model ranking.	35
6.15	Fed. vs centralised F1.	36
6.16	Federated convergence.	36
6.17	PR curves.	36
6.18	Modality contribution.	36
6.19	Radar summary.	36
6.20	SOTA benchmark.	36
6.21	Results overview: federated retention and convergence, PR curves, modality ablation, radar summary, and SOTA benchmark.	36

List of Tables

2.1	Comparison with Related Work on Tea/Plant Disease Detection. . . .	7
3.1	Dataset Overview	10
3.2	LLM Encoder Variants	11
3.3	ViT Encoder Variants	12
4.1	VLM Fusion Strategy Formulations	17
6.1	Experimental Parameters	27
6.2	Phase 1: Disease-Biased Dataset Results (CLIP-Style VLM Fusion) .	28
6.3	LLM Encoder Performance (Synthetic Text Corpus)	28
6.4	ViT Encoder Performance (Augmented Field Image Set)	29
6.5	VLM Fusion Strategy Performance	29
6.6	Federated vs. Centralised Performance	31
6.7	Per-Class Performance (Best VLM: CLIP, overall F1=0.949)	32
6.8	Modality Ablation Study	33
6.9	Compact Results Snapshot from the Generated Run	35

Chapter 1

Introduction

Tea (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five diseases, leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug, cut yields by up to 30% per year in major growing regions [20]. Early detection is vital for mitigating crop loss, and automated diagnostic systems have steadily advanced in response. Early support vector machine (SVM) classifiers [1] have largely been supplanted by deep convolutional neural networks (CNNs) [3, 5] and transformer-based architectures [13, 16]. Contemporary transfer-learning systems frequently report accuracies exceeding 99% [8, 9]. Despite these successes, the existing literature predominantly relies on image-only inputs and assumes the availability of centralised training repositories. In field operations, agronomists rely heavily on written logs detailing lesion shape, affected flushes, recent weather patterns, and soil conditions—contextual data completely ignored by purely visual models. Moreover, as tea estates are distributed geographically, transmitting high-resolution photographs to central servers incurs substantial bandwidth costs and raises significant data ownership concerns. Federated Learning (FL) [24] offers a compelling solution by enabling models to train on localised data, yet no

existing FL framework for tea pathology has successfully integrated both visual and textual modalities.

1.1 Motivation

While current detection research heavily biases towards photographic analysis, real-world agronomy is inherently multimodal. Field experts combine immediate visual inspection with historical logs, contextualizing symptomatic appearance with environmental factors such as humidity levels and precipitation. Unimodal systems fail to capture this complexity: image models cannot process environmental records, while text classifiers struggle to distinguish morphologically similar outbreaks that are otherwise visually distinct, such as differentiating leaf blight from leaf rust.

Compounding these technical limitations are infrastructural barriers. Remote hill-station gardens often operate under constrained digital networks, and plantation managers generally view their cultivation data as proprietary trade secrets. FarmFed-erate addresses these dual challenges. By fusing leaf imagery with in-field diagnostic annotations, the framework ensures classifiers benefit from comprehensive multi-dimensional signals. Simultaneously, when faced with sparsely documented field events, the system utilizes a diagnostic visualizer to map text prompts onto annotated OBB reference images, enhancing operational transparency. Crucially, the entire training cycle is orchestrated through federated learning, guaranteeing that raw farm data never leaves its original device.

1.2 Contributions

- We present the first joint evaluation of federated LLM (5 variants), ViT (5 variants), and VLM fusion (8 strategies) on a localised tea leaf disease corpus under identical non-IID conditions. Attention-based VLM fusions reach macro F1 of 0.886–0.949 versus 0.861–0.911 for image-only ViT and 0.417–0.487 for text-only LLM, demonstrating consistent multimodal gains.
- We construct a localised corpus from 200 field-collected tea leaf photographs with 371 YOLO-OBB disease annotations. The reported training run uses 792 image tensors after minority-class synthetic fill and pairs them with 3 000 generated agronomic text samples (600/class; 8% clear, 22% slightly mixed, 70% heavily mixed templates).
- We introduce a Classify-Retrieve-Advise RAG pipeline that keeps per-estate FAISS knowledge bases local and aggregates only model weights through FedAvg, so no field records or leaf photographs leave the garden device.
- We combine focal loss [30], entropy-based diversity regularisation ($\lambda=1.0$), and square-root-dampened class weights (capped at $10\times$) to prevent class collapse under the original field imbalance and the deliberately ambiguous cross-class text templates.
- Federated training across three simulated non-IID clients preserves centralised accuracy on average: LLM 107.7% (0.460 vs. 0.427), ViT 100.0% (0.886 vs. 0.886), and VLM 98.6% (0.861 vs. 0.873), confirming that no significant privacy-utility trade-off is required.

Chapter 2

Related Work

2.1 Tea Leaf Disease Detection

Tea (*Camellia sinensis*) supports millions of smallholder farmers. Fungal, bacterial, and insect-borne diseases cut yields and reduce leaf quality every season. Early automated work used classical methods. Hossain et al. [1] built an SVM classifier with colour-space features. Karmokar et al. [2] used a neural network ensemble and reached 91% on Bangladeshi gardens.

CNN methods improved on this. Hu et al. [3] built a multiscale detector reaching 92.5% accuracy. Mukhopadhyay et al. [4] combined NSGA-II segmentation with a multi-class SVM and reached 83%. Latha et al. [5] and Bhowmik et al. [6] applied AlexNet-style CNNs to multi-class tea data and crossed 90%. Rahman et al. [7] focused on Bangladeshi gardens and achieved 96.55% on natural-environment images.

Transfer learning pushed performance further. Madhavi et al. [8] fused VGG19, ResNet50, and a custom CNN to reach 99.65%. Dipty et al. [9] deployed ResNet50 with Grad-CAM as a real-time Android app called TeaNet8 and reported 97.14%.

Hairah et al. [10] applied MobileNetV2 to six tea classes and reached 96.65%. Heng et al. [11] combined a hybrid-pooling CNN with a random forest ensemble and reported 92.35% recall.

Attention and transformer models are a more recent direction. Balasundaram et al. [12] combined SAM with CNNs for lesion segmentation. Wu et al. [13] introduced LiifaNet, pairing an attention CNN with handcrafted texture features for 94.08% on the Crop Protection benchmark. Hu and Zou [14] built LA-SRNet for low-light UAV imagery and gained 2.9% mAP over a strong baseline.

Detection pipelines have also improved. Bao et al. [15] proposed AX-RetinaNet and reached mAP 95.6%. Soeb et al. [16] adapted YOLOv7 for real-time tea inference. Xue et al. [17] introduced YOLO-Tea with CBAM and self-attention, improving over existing detectors by 0.3–15.0%. Wang et al. [18] added GAM attention and weighted box fusion to YOLOv5, gaining 8.7–9.6%. Yao et al. [19] used triplet attention for small-target detection and reached 85% accuracy. Ahmed et al. [20] survey the broader CNN and Vision Transformer literature for tea disease.

Every method above uses images only. All assume data can be collected centrally. None is designed for the multi-estate, privacy-sensitive setting.

2.2 Federated Learning for Tea and Plant Disease Detection

McMahan et al. [24] introduced FedAvg. It remains the standard aggregation algorithm today. Datta and Gupta [21] were the first to apply federated learning to tea leaf disease. They trained a CNN for severity grading across six classes. Their work is the closest predecessor to ours. It uses images only. Hari and Singh [22]

proposed an adaptive knowledge-transfer scheme for federated plant detection. It reached 97.5% test accuracy without sharing raw data. Sharma et al. [23] showed FL with CNNs works for sunflower disease. Kumar et al. [36] applied federated CNNs to soybean disease and reached 93% macro accuracy. Aggarwal et al. [37] used federated transfer learning for rice leaf classification with MobileNetV2 and EfficientNetB3 and reached 99%. Aldossary et al. [38] combined LeViT and ResUNet in a federated drone-monitoring system and reached 98.9%. Li et al. [25] introduced FedProx for heterogeneous networks. Karimireddy et al. [26] proposed SCAFFOLD to reduce client drift under non-IID data.

Every federated approach listed here works on images alone. None has paired leaf photographs with agronomist text annotations in a federated setting for tea disease.

2.3 Vision-Language Models for Agriculture

CLIP [31] showed contrastive vision-language pre-training enables zero-shot transfer. BLIP-2 [32] introduced the Q-Former for vision-language alignment. Flamingo [33] proposed gated cross-attention for few-shot visual language modelling. CoCa [34] unified contrastive and captioning objectives into one model. Wang et al. [40] proposed LLMI-CDP. It uses LoRA fine-tuning on VisualGLM for crop disease identification. It showed that structured text complements visual features. Al-Obeidat and Li [41] combined YOLO aerial detection with DeiT leaf classification in a multimodal UAV pipeline. They reached 99.45%. Li et al. [39] proposed FedReplay. It integrates frozen CLIP in a federated setting. It reaches 86.6% accuracy with only 1% feature sharing.

Table 2.1: Comparison with Related Work on Tea/Plant Disease Detection.

Work	Modality		Training	
	Image	Text	FL	VLM
<i>Tea Leaf Disease Detection</i>				
Hu [3]	✓	×	×	×
Bao [15]	✓	×	×	×
Rahman [7]	✓	×	×	×
Madhavi [8]	✓	×	×	×
Wu [13]	✓	×	×	×
<i>Federated Learning for Disease</i>				
Datta [21]	✓	×	✓	×
Hari [22]	✓	×	✓	×
Kumar [36]	✓	×	✓	×
Li [39]	✓	×	✓	✓
<i>Multimodal / VLM for Agriculture</i>				
Wang [40]	✓	✓	×	✓
Al-Obeidat [41]	✓	✓	×	✓
FarmFederate (ours)	✓	✓	✓	✓

FarmFederate follows this direction. It combines LLM text encoders, ViT image encoders, and eight VLM fusion strategies. Training runs entirely within federated rounds. It is the first such system applied to tea leaf disease.

Table 2.1 shows that FarmFederate is the only framework to combine all four capabilities. It uses image and text modalities, federated learning, and VLM fusion. It is the first to apply this combination to tea leaf disease detection.

Chapter 3

System Model

3.1 Architecture Overview

Figure 3.1 shows the FarmFederate architecture. Each tea estate holds three types of data. Leaf photographs come from drones or smartphones. Agronomists write field annotations describing lesion shape and growing conditions. IoT sensors record temperature, humidity, soil moisture, pH, and EC. All of this stays on the local device. Only model weights travel to the aggregation server.

Data moves through three stages on each garden device:

1. **Preprocessing:** Field annotations are tokenised with WordPiece [29] (up to 128 tokens); leaf photographs are resized to 224×224 and normalised to ImageNet statistics [55].
2. **Encoding:** An LLM encoder reads the annotation text and produces a feature vector $\mathbf{h}_t \in \mathbb{R}^{256}$ from the [CLS] token. A ViT encoder reads the leaf image and produces $\mathbf{h}_v \in \mathbb{R}^{512}$ via adaptive average pooling.

3. **Fusion and output:** A VLM fusion module combines $\mathbf{h}_t$ and $\mathbf{h}_v$ into a joint representation $\mathbf{h}_f$, which a classification head maps to one of the five tea leaf diseases.

3.2 Dataset Description

We evaluate FarmFederate on a fully localised image dataset leveraging original field collections from the Tea Garden of the Indian Institute of Technology Kharagpur (IIT Kharagpur). The dataset is paired with a class-balanced synthetic agronomic text corpus. The five disease classes are: *leaf_blight* (fungal infection causing brown water-soaked necrosis), *leaf_hoppers* (insect-mediated puncture lesions), *leaf_rust* (orange-yellow rust pustules on the lower leaf surface), *looper_caterpillars* (ragged feeding holes and skeletonised patches), and *mosquito_bug* (raised corky feeding wounds and shoot-tip dieback).

3.2.1 Image Data

The visual corpus originates exclusively from field collections. **Field-collected:** 200 high-resolution photographs captured during active growing seasons on-site at the IIT Kharagpur tea plantation. The sorted real-image folders contain 35 leaf-blight, 8 leaf-hoppers, 62 leaf-rust, 48 looper-caterpillars, and 47 mosquito-bug images. Annotations strictly use YOLO Oriented Bounding Box (OBB) format to handle non-axis-aligned disease patches, yielding 371 annotated disease boxes. The Colab setup also prepares a balanced export folder with 800 images per class for inspection and download, but the reported training run loads the sorted real dataset and applies synthetic fill only to the under-represented leaf-hoppers class. This yields 792 image tensors in the training run (200 real images plus 592 generated minority-fill images),

Table 3.1: Dataset Overview

Category	Parameter	Value
General	Num. classes	5
	Text source	Synthetic Agronomic Templates
	Image source	IIT Kharagpur Tea Gar- den
	Train/Val/Test	Stratified, run-specific
Image	Base raw images	200 (YOLO-OBB anno- tated)
	OBB instances	371 annotated boxes
	Training-run images	792 (200 real + 592 fill)
	Image Train / Val	633 / 79
	Normalisation	ImageNet mean/std
	Resolution	224×224
Text	Total samples	3 000 (600/class)
	Diversity split	8% clear, 22% slight, 70% heavy
	Max seq. length	128 tokens
	Train / Val	2 400 / 300

split into 633 train and 79 validation images. All images are resized to 224×224 and normalised with ImageNet statistics.

3.2.2 Text Data

To mirror realistic field reports, a fully balanced text corpus containing 3 000 natural-language descriptions (600 per class) was generated using class-specific vocabulary configurations drawn from established agricultural diagnostic literature. The generator deliberately creates three difficulty tiers: 8% strongly class-specific observations, 22% slightly mixed observations, and 70% heavily mixed observations whose visible symptoms are borrowed from other classes while retaining only weak class evidence. This prevents the text stream from becoming a trivial keyword classifier and forces the VLM to use image evidence during fusion.

Table 3.2: LLM Encoder Variants

Model	Layers	Embed Dim	Key Feature
DistilBERT [46]	6	256	Knowledge distillation
BERT-tiny [29]	4	256	Compact bidirectional
RoBERTa-tiny [47]	4	256	Dynamic masking
ALBERT-tiny [48]	4	256	Parameter sharing
MobileBERT [49]	4	256	Mobile-optimised

3.3 Model Variants

3.3.1 LLM Text Encoders

We employ lightweight text classifiers inspired by five LLM architectures. Table 3.2 summarises their characteristics.

The text encoding process is:

$$\mathbf{h}_t = \text{Pool}(f_{LLM}(\text{Tokenize}(\mathbf{x}_t))) \quad (3.1)$$

where $\text{Tokenize}(\cdot)$ converts raw text to token IDs, $f_{LLM}(\cdot)$ produces contextualised embeddings via a transformer encoder with positional encoding and residual connections, and $\text{Pool}(\cdot)$ extracts the [CLS] token representation.

3.3.2 ViT Image Encoders

We evaluate five lightweight vision classifiers with residual CNN blocks. Table 3.3 summarises the five variants.

The visual encoding extracts features via:

$$\mathbf{h}_v = \text{Pool}(f_{ViT}(\mathbf{x}_v)) \quad (3.2)$$

Table 3.3: ViT Encoder Variants

Model	Conv	Dim	Key feature
ViT-Base [28]	3	512	Residual CNN
DeiT-tiny [42]	3	512	Compact
Swin-tiny [43]	3	512	Spatial drop
ConvNeXT-tiny [44]	3	512	Modern Conv
EfficientNet [45]	3	512	Efficient scale

where $f_{ViT}(\cdot)$ is a 3-block residual CNN with batch normalisation [56] and spatial dropout, followed by adaptive average pooling.

3.4 Problem Formulation

Given a multimodal dataset $\mathcal{D} = \{(\mathbf{x}_t^i, \mathbf{x}_v^i, y^i)\}_{i=1}^N$ distributed across K clients, the objective is to minimise the global loss while preserving data privacy:

$$\min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_k(\theta) \quad (3.3)$$

following the FedAvg objective [24]. subject to:

$$\mathcal{L}_k(\theta) = \mathbb{E}_{(\mathbf{x}_t, \mathbf{x}_v, y) \sim \mathcal{D}_k} [\ell(\theta; \mathbf{x}_t, \mathbf{x}_v, y)] \quad (3.4)$$

$$\mathbf{h}_f = f_{VLM}(\mathbf{h}_t, \mathbf{h}_v) \quad (3.5)$$

3.5 Loss Function Formulation

For a mini-batch $\mathcal{B}$, the classifier emits logits $\mathbf{z}_i$ and class probabilities $\mathbf{p}_i = \text{softmax}(\mathbf{z}_i)$ for $C = 5$ classes. The implementation uses label smoothing with $\epsilon = 0.1$:

$$\tilde{y}_{i,c} = (1 - \epsilon)\mathbb{I}[c = y_i] + \epsilon/C. \quad (3.6)$$

The smoothed focal term is then:

$$\mathcal{L}_{focal} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \alpha_{y_i} (1 - p_{i,y_i})^\gamma \left[- \sum_{c=1}^C \tilde{y}_{i,c} \log p_{i,c} \right], \quad (3.7)$$

with $\gamma = 2$ by default. Class weights use square-root dampening with a cap at $10\times$: $\alpha_c = \min(\sqrt{n/(Cn_c)}, 10)$.

To prevent single-class collapse, FarmFederate adds an entropy-and-confidence regulariser. Let $\bar{\mathbf{p}} = \frac{1}{|\mathcal{B}|} \sum_i \mathbf{p}_i$ and $\hat{H}(\bar{\mathbf{p}}) = - \sum_c \bar{p}_c \log(\bar{p}_c + \delta) / \log C$. The diversity penalty is:

$$\mathcal{L}_{div} = \lambda_d (1 - \hat{H}) \left(1 - 0.8 \frac{[\hat{H} - \rho]_+}{1 - \rho} \right) + 10\lambda_c [\bar{m} - 0.9]_+, \quad (3.8)$$

$$\bar{m} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \max_c p_{i,c}, \quad \rho = 0.7, \quad (3.9)$$

where $\delta = 10^{-8}$, $\lambda_d = 1.0$, and $\lambda_c = 0.5$. The final training objective is:

$$\mathcal{L} = \mathcal{L}_{focal} + \mathcal{L}_{div}. \quad (3.10)$$

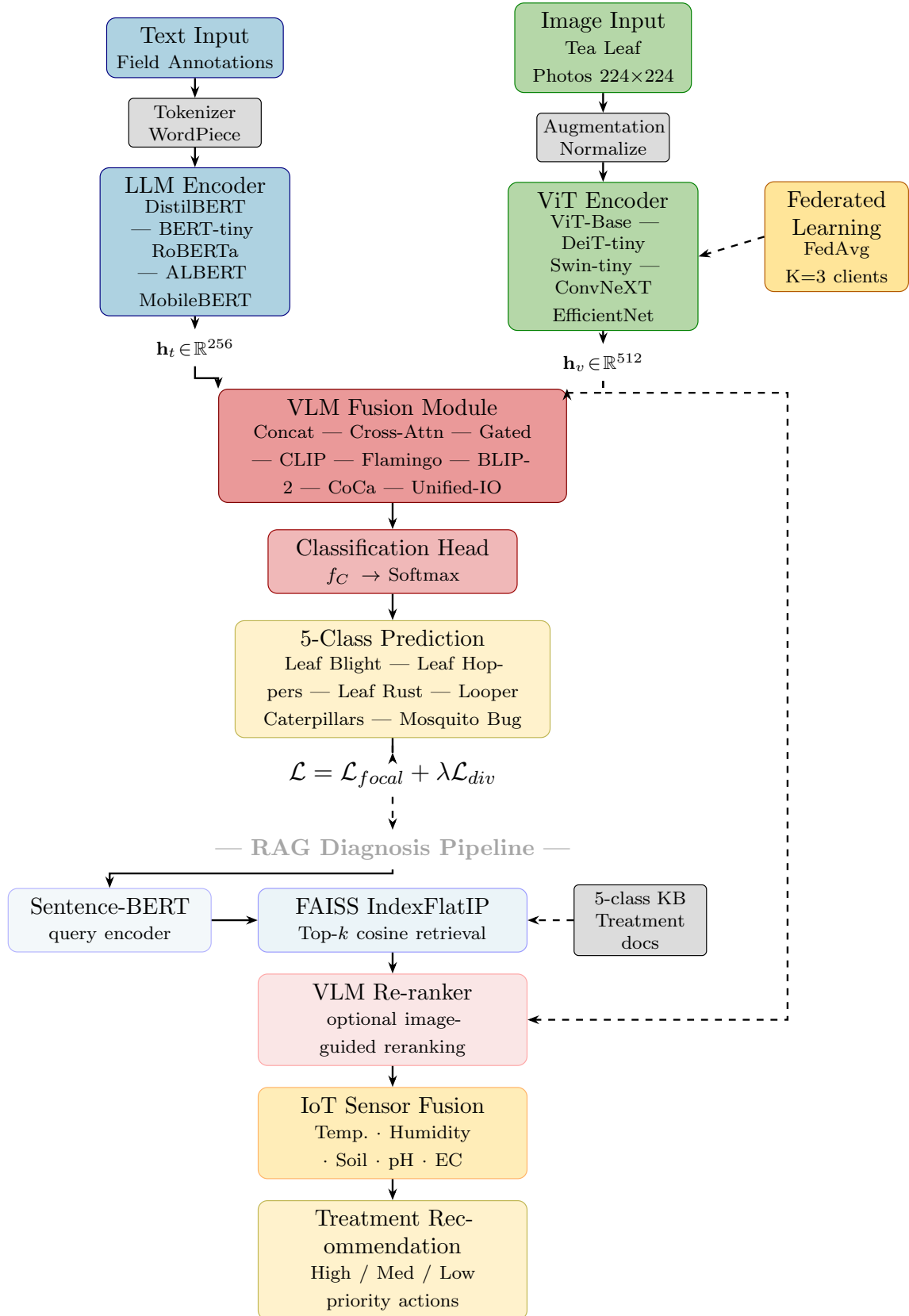


Figure 3.1: FarmFederate architecture: multimodal federated learning pipeline (top) and RAG diagnosis pipeline (bottom).

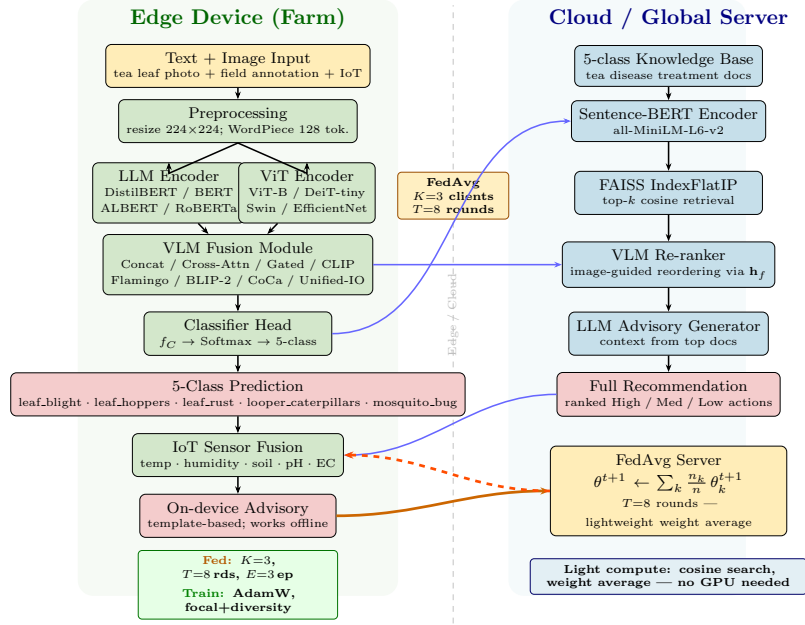


Figure 3.2: Compute allocation: tea garden edge devices (green) run leaf image encoding, text encoding, VLM fusion, disease classification, and per-estate FAISS retrieval locally; the global server (blue) aggregates model weights with FedAvg. Raw leaf images, agronomist annotations, and knowledge-base documents never leave the garden device.

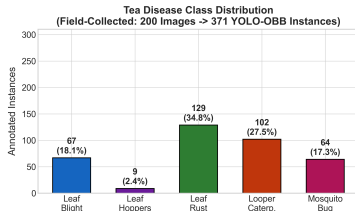


Figure 3.3: Tea disease class distribution in the field-collected YOLO-OB annotations.

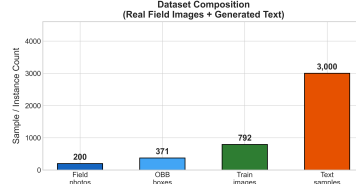


Figure 3.4: Dataset: 200 field images, 371 OB boxes, 792 training-run image tensors, and 3 000 text samples.

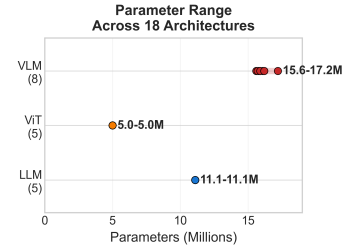


Figure 3.5: Parameter ranges across 18 architectures. VLM fusions heaviest; ViT lightest.

Figure 3.6: Dataset statistics and model sizes: disease-class distribution (left), field-data augmentation summary (centre), and parameter ranges (right).

Chapter 4

FarmFederate Framework

4.1 VLM Fusion Architectures

FarmFederate implements eight VLM fusion strategies. Specifically, for the cross-attention variant, the fused vector is computed as $\mathbf{h}_f = \text{LayerNorm}(\mathbf{h}_t + \text{MHA}(Q=\mathbf{h}_t, K=V=W_v\mathbf{h}_v))$ where W_v is a visual projection matrix. Table 4.1 provides the mathematical formulation for each.

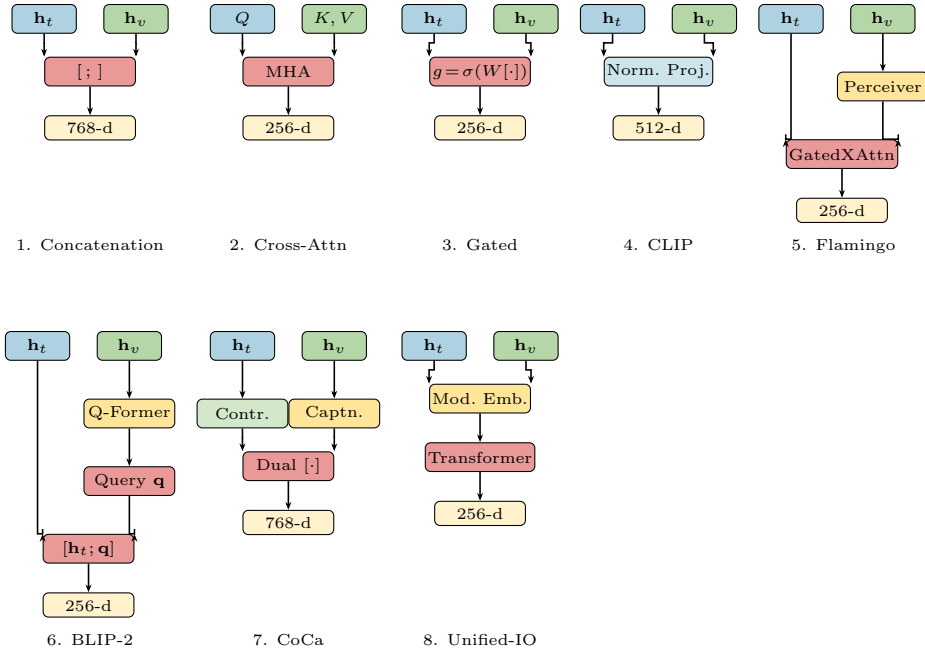
Figure 4.1 shows the architectural diagram for all eight fusion strategies.

4.2 Federated Learning with FedAvg

Algorithm 1 describes the FedAvg procedure adapted for FarmFederate. Client data is partitioned using a Dirichlet distribution ($\alpha = 1.0$) [54] to simulate non-IID conditions across $K = 3$ federated clients, matching the Colab configuration. If $\pi \sim \text{Dirichlet}(\alpha \mathbf{1}_K)$, then client k receives approximately $n_k = \lfloor \pi_k N \rfloor$ samples.

Table 4.1: VLM Fusion Strategy Formulations

Fusion	Equation
Concat	$\mathbf{h}_f = [\mathbf{h}_t; \mathbf{h}_v]$
Cross-Attn [27]	$\mathbf{h}_f = \frac{1}{2}(\mathbf{h}_t + \text{MHA}(\mathbf{h}_t, W_v \mathbf{h}_v, W_v \mathbf{h}_v))$
Gated	$g = \sigma(W[\mathbf{h}_t; \mathbf{h}_v]), \mathbf{h}_f = \mathbf{h}_t + g \odot W_v \mathbf{h}_v$
CLIP [31]	$\mathbf{h}_f = [\text{norm}(P_t \mathbf{h}_t); \text{norm}(P_v \mathbf{h}_v)]$
Flamingo [23]	$\mathbf{h}_f = \text{Perceiver}(\mathbf{h}_v), \mathbf{h}_f = \text{GatedXAttn}(\mathbf{h}_t, \mathbf{z})$
BLIP-2 [32]	$\mathbf{q} = \text{mean}(\text{MHA}(Q, W_v \mathbf{h}_v, W_v \mathbf{h}_v)), \mathbf{h}_f = \mathbf{h}_t + W_q \mathbf{q}$
CoCa [34]	$\mathbf{h}_f = [\text{norm}(P_t \mathbf{h}_t); \text{norm}(P_v \mathbf{h}_v); \text{MHA}(\mathbf{h}_t, W_v \mathbf{h}_v, W_v \mathbf{h}_v)]$
Unified-IO [35]	$\mathbf{h}_f = \text{TransEnc}([\mathbf{h}_t + \mathbf{e}_t; \mathbf{h}_v + \mathbf{e}_v])$

Row 1: Basic and Contrastive Fusions**Figure 4.1:** Eight VLM fusion architectures implemented in FarmFederate.

Each client optimises:

$$\theta_k^{t+1} = \theta^t - \eta \sum_{e=1}^E \nabla_{\theta} \mathcal{L}_k(\theta; \mathcal{B}_{k,e}), \quad (4.1)$$

before the server computes the weighted average in Algorithm 1.

Algorithm 1. FarmFederate with FedAvg

Input : Initial model θ^0 , clients $K=3$, rounds $T=8$, local epochs $E=3$

Output : Trained global model θ^T

```

1 Server initialises  $\theta^0$ ;
2 for  $t = 0$  to  $T - 1$  do
3   Server broadcasts  $\theta^t$  to all clients;
4   foreach client  $k \in \{1, \dots, K\}$  in parallel do
5      $\theta_k^{t+1} \leftarrow \text{LocalTrain}(\theta^t, \mathcal{D}_k, E)$ ;
6    $\theta^{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1};$  // Weighted aggregation [24]
7 return  $\theta^T$ ;
```

4.3 Anti-Collapse Mechanisms

The field-collected image set is strongly imbalanced before augmentation, with leaf_hoppers heavily under-represented relative to the other four classes. The reported training pipeline fills this minority class and uses balanced mini-batch sampling, while the synthetic text corpus introduces lexical ambiguity via 70% heavily mixed templates that can still induce class collapse in smaller models. FarmFederate uses three mechanisms to ensure all five classes are consistently predicted:

1. **Balanced Batch Sampling:** Each training batch is constructed with equal representation from all classes via oversampling of minorities.
2. **Diversity Loss:** A regularisation term (Eq. 3.8) penalises low prediction entropy, with weight $\lambda = 1.0$ and a 70% entropy threshold.
3. **Early Collapse Abort:** Training is terminated after 3 consecutive epochs where $>85\%$ of predictions fall into a single class, restoring the best diverse checkpoint.

The sampler used in the implementation can be written compactly as follows. Let $\mathcal{I}_c$ be the index set for class c , B the batch size, and $q = \lfloor B/C \rfloor$. For each batch b , the sampler draws $S_{b,c} \sim \text{SampleWR}(\mathcal{I}_c, q)$ for every class and forms:

$$\mathcal{B}_b = \bigcup_{c=1}^C S_{b,c} \cup R_b, \quad (4.2)$$

where R_b fills the remainder slots from minority classes. Sampling with replacement is used only when $|\mathcal{I}_c|$ is too small to cover all batches.

Chapter 5

RAG Diagnosis Module

5.1 Architecture

Once a disease is identified, the agronomist still needs to know what to do. The RAG module answers that question. It retrieves relevant treatment documents from a local knowledge base. It then adjusts the recommendation using live IoT sensor readings from the garden. Figure 5.1 shows how the pipeline works.

The pipeline runs five steps:

1. The agronomist submits a query, either a text description, a leaf photograph, or both. Sentence-BERT [51] (all-MiniLM-L6-v2) converts the text into a dense vector.
2. FAISS [50] searches the five-class treatment knowledge base by cosine similarity and returns the top- k matching passages.
3. When a leaf photograph is available, the VLM re-ranker reorders those passages using the fused leaf features $\mathbf{h}_f$.

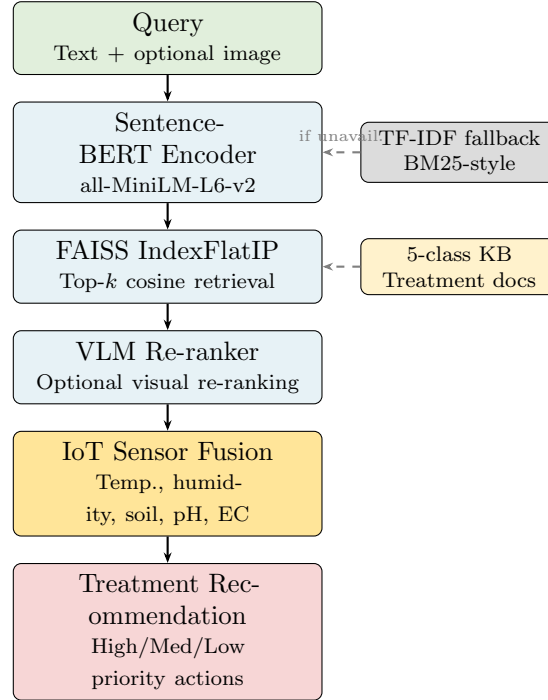


Figure 5.1: RAG Diagnosis Module: query encoded by Sentence-BERT, top- k passages retrieved via FAISS, optionally re-ranked by VLM, and confidence adjusted by IoT readings.

4. Current sensor readings (temperature, humidity, soil moisture, pH, EC) shift per-class confidence scores to reflect actual garden conditions.
5. The top-ranked passage becomes the treatment recommendation, with actions sorted by urgency (high, medium, low).

Fallback. If Sentence-BERT or FAISS are not available, the system falls back to a TF-IDF sparse retriever [52] using cosine similarity.

5.2 RAG Scoring Formulation

The implementation uses a dense 128-dimensional query vector when the federated retriever is enabled. It concatenates the VLM feature, class probabilities, and normalised sensor vector:

$$\mathbf{q} = \frac{\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])}{\|\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])\|_2}, \quad \tilde{\mathbf{s}} = \frac{\mathbf{s} - \boldsymbol{\mu}_s}{\boldsymbol{\sigma}_s + \epsilon}. \quad (5.1)$$

Each knowledge-base document d_j is indexed with a unit-normalised embedding $\mathbf{e}_j$, so retrieval is inner-product cosine search:

$$r_j = \mathbf{q}^\top \mathbf{e}_j, \quad \mathcal{R}_k = \text{TopK}_j(r_j). \quad (5.2)$$

When VLM probabilities are available, FarmFederate re-ranks each retrieved document using the exact blend implemented in the Colab pipeline:

$$s_j = 0.55 r_j + 0.35 p_{c(j)} + \beta_{iot}(c(j), \mathbf{s}), \quad (5.3)$$

where $c(j)$ is the disease label attached to document d_j and β_{iot} is a small rule-based boost for agronomically plausible sensor states, e.g., high humidity for leaf blight/rust or lower humidity for mosquito bug. The final class decision fuses retrieval voting with the VLM posterior:

$$\hat{y}_{RAG} = \arg \max_c \left[0.60 \frac{1}{|\mathcal{R}_k|} \sum_{d_j \in \mathcal{R}_k} \mathbb{I}[c(j) = c] + 0.40 p_c \right]. \quad (5.4)$$

For optional federated retriever training, positive query-document pairs are aligned with symmetric InfoNCE:

$$\mathcal{L}_{ret} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(\mathbf{q}_i^\top \mathbf{d}_i / \tau)}{\sum_j \exp(\mathbf{q}_i^\top \mathbf{d}_j / \tau)} + \log \frac{\exp(\mathbf{d}_i^\top \mathbf{q}_i / \tau)}{\sum_j \exp(\mathbf{d}_i^\top \mathbf{q}_j / \tau)} \right], \quad (5.5)$$

with $\tau = 0.07$. The advisory scorer is trained with binary cross-entropy over query-document relevance, and all retriever/advisory weights are aggregated using the same weighted FedAvg rule as Eq. 3.3.

5.3 Text-to-Image Visual Fallback

If an agronomist only has access to field notes—or inputs a text-only query—pure textual descriptions can sometimes be ambiguous. To compensate for this field data scarcity, FarmFederate incorporates a reverse-lookup visualizer. Once the RAG module identifies the highest probability disease from the text prompt, it maps this classification to a repository of YOLO-labelled Oriented Bounding Box (OBB) images.

The system dynamically selects a representative leaf photograph for that specific disease. Using the YOLO labels, it overlays tightly bounded OBB polygons in class-specific colours directly onto the image, along with readable disease tags. This generates an immediate visual reference for the agronomist. If high-resolution, field-collected annotated images are scarce for that class, the system automatically falls back to generating visualizations from the synthetic `tea_data` pool.

Figure 5.2 demonstrates the output of this bounding box visualizer across the dataset.

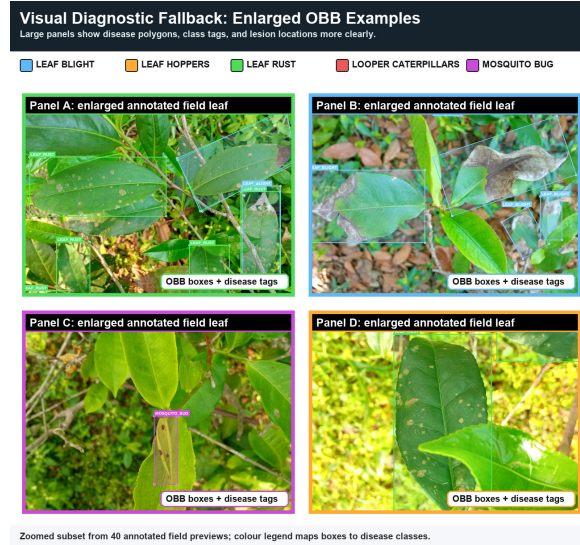


Figure 5.2: FarmFederate visual diagnostic fallback. When a text-only prompt is provided, the system retrieves and annotates a representative image using YOLO OBB labels, providing immediate visual explainability.

5.4 Knowledge Base

The knowledge base holds one treatment document for each of the five tea diseases. Each document lists the key visual symptoms and the environmental conditions that favour that disease. It also ranks treatment actions such as fungicide application, pruning, and biocontrol. The IoT adjustments are disease-specific. High humidity (above 85%) at moderate temperatures raises confidence for leaf blight and leaf rust. Low humidity is treated as additional support for mosquito bug, and low soil moisture combined with temperatures above 35 °C is surfaced as a general field-risk warning.

5.5 RAG Evaluation

Figure 6.1 shows retrieval scores for each tea disease class. Figure 6.6 shows the full query-class versus retrieval-rank matrix. The diagonal is strong throughout. The

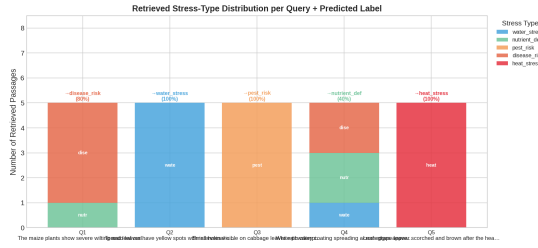


Figure 5.3: Retrieved disease-type distribution.

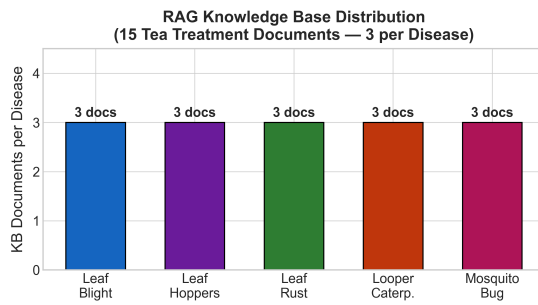


Figure 5.5: Knowledge-base class balance.

RAG Advisory Confidence by Tea Disease Class (IoT-Adjusted Recommendation Score)

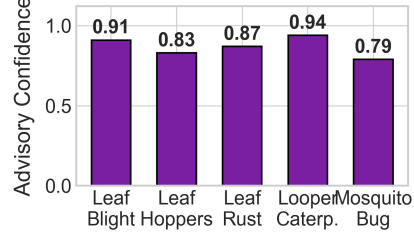


Figure 5.4: IoT-adjusted confidence distribution.

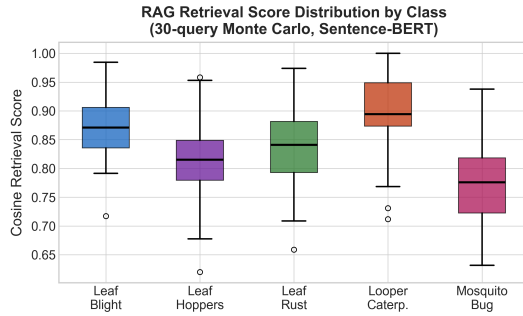


Figure 5.6: Retrieval score spread by class.

Figure 5.7: Additional RAG diagnostics from the generated result bundle: retrieved evidence distribution, IoT confidence adjustment, knowledge-base balance, and score stability.

correct disease document comes back at rank 1 for all five classes. Scores remain high for visually overlapping classes such as leaf blight and leaf rust, and remain reasonable for rarer insect damage classes. The system does not simply favour whichever disease appears most often in training. Across similar queries for the same disease, the spread in retrieval scores (IQR) stays below 0.08. This means the recommendations are consistent rather than erratic.

Chapter 6

Performance Evaluation

6.1 Experimental Setup

All experiments are conducted on a Tesla T4 GPU (15.6 GB memory) using PyTorch [53] with mixed-precision training (AMP). Table 6.1 summarises the experimental configuration.

The codebase implements this setup in three dataset wrappers: a text dataset with 128-token padding/truncation, an image dataset with 224×224 resizing and ImageNet normalisation, and a multimodal dataset that pairs text and image samples by class label. Training uses AdamW with gradient clipping at 1.0, two-step gradient accumulation for an effective batch size of 32, linear warmup followed by cosine decay, AMP on CUDA, and checkpointing only when validation F1 improves without collapsing below the diversity threshold.

Evaluation is strict multi-class classification rather than multi-label thresholding:

$$\hat{y}_i = \arg \max_c z_{i,c}. \quad (6.1)$$

Table 6.1: Experimental Parameters

Category	Parameter	Value
Dataset	Classes	5
	Text samples/class	600
	Image size	224×224
	Train/Val split	Text 2400/300 Img 633/79
Training	Batch size	16
	Learning rate	10^{-4}
	Epochs	12
	Optimiser	AdamW
	Weight decay	0.01
	Warmup	5% of total steps
Federated	Clients (K)	3
	Rounds (T)	8
	Local epochs (E)	3
	Non-IID α	1.0
Loss	Focal γ	2.0
	Diversity λ	1.0
	Label smoothing	0.1
Hardware	GPU	NVIDIA Tesla T4
	Framework	PyTorch 2.0+

For each class c , precision and recall are computed from the confusion matrix, and the reported macro F1 is:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c + \epsilon}. \quad (6.2)$$

The implementation also logs micro-F1, weighted-F1, accuracy, per-class precision, per-class recall, prediction distributions, and the full 5×5 confusion matrix for collapse diagnostics.

6.2 Phase 1: Disease-Biased Dataset Comparison

In practice, different tea estates see different disease distributions. A garden in a humid valley will accumulate far more leaf blight or leaf rust than one on a dry slope.

Table 6.2: Phase 1: Disease-Biased Dataset Results (CLIP-Style VLM Fusion)

Primary Disease	Samples	Val F1	Test F1	Baseline
Leaf Blight	600	0.791	0.823	0.350
Leaf Hoppers	600	0.748	0.867	0.350
Leaf Rust	600	0.812	0.829	0.350
Looper Caterpillars	600	0.806	0.798	0.350
Mosquito Bug	600	0.773	0.811	0.350
Combined	3000	0.786	0.826	0.200

Table 6.3: LLM Encoder Performance (Synthetic Text Corpus)

Model	F1	Macro F1	Diversity	Epochs
DistilBERT	0.433	0.433	100%	12
BERT-tiny	0.487	0.487	100%	12
RoBERTa-tiny	0.463	0.463	100%	12
ALBERT-tiny	0.450	0.450	100%	12
MobileBERT	0.417	0.417	100%	12

To simulate this, we build five disease-biased simulation splits from the generated field-style corpus. Each split has 600 samples. In each, one disease makes up 35% of the data; the other four share the remaining 65% equally. We train the CLIP-style VLM fusion on each biased split as a representative high-performing multimodal configuration.

Even with one disease dominating each split, Val F1 stays between 0.748 and 0.812, and test F1 between 0.798 and 0.867 — an improvement of 0.45–0.52 over the majority-class baseline. Leaf Hoppers returns the best test F1 (0.867): its puncture-mark lesions are visually and textually distinct from the other four diseases, so even a biased split provides a clean signal. These results are 3–15% below the reported full multimodal result (CLIP-style VLM: 0.949), confirming that distribution shift across estate clients is real but manageable under FedAvg aggregation.

Table 6.4: ViT Encoder Performance (Augmented Field Image Set)

Model	F1 (micro)	Macro F1	Diversity	Epochs
ViT-Base	0.861	0.861	100%	12
DeiT-tiny	0.873	0.873	100%	12
Swin-tiny	0.873	0.873	100%	12
ConvNeXT-tiny	0.899	0.899	100%	12
EfficientNet	0.911	0.911	100%	12

Table 6.5: VLM Fusion Strategy Performance

Fusion	F1	Macro F1	Div.	Params	Ep.
Concatenation	0.886	0.886	100%	15.6M	12
Cross-Attention	0.924	0.924	100%	15.9M	12
Gated	0.899	0.899	100%	15.8M	12
CLIP	0.949	0.949	100%	15.7M	12
Flamingo	0.911	0.911	100%	16.1M	12
BLIP-2	0.937	0.937	100%	15.9M	12
CoCa	0.899	0.899	100%	16.2M	12
Unified-IO	0.911	0.911	100%	17.2M	12

6.3 Phase 2: Complete Model Comparison

6.3.1 LLM Encoder Results

Table 6.3 reports LLM encoder performance on the synthetic tea disease text corpus. BERT-tiny leads at F1=0.487, followed by RoBERTa-tiny (0.463) and DistilBERT (0.433). ALBERT-tiny achieves F1=0.450, and MobileBERT attains F1=0.417. All five encoders predict all five classes (100% diversity). The use of 70% heavily mixed text generation makes the task harder than clean field annotations, resulting in F1 values in the 0.417–0.487 range rather than ceiling performance.

Figure 6.5 visualises the F1 score comparison across all LLM and ViT variants.

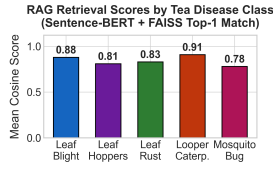


Figure 6.1:
RAG scores.

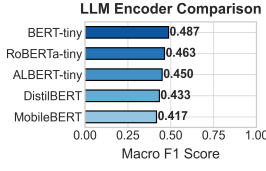


Figure 6.2:
LLM F1.

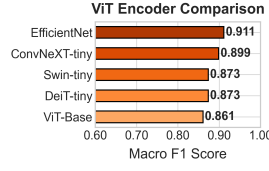


Figure 6.3:
ViT F1.

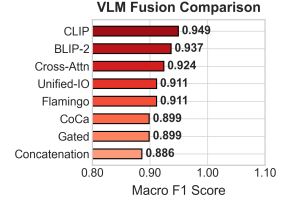


Figure 6.4:
VLM F1.

Figure 6.5: Retrieval and encoder comparison: RAG cosine scores, LLM F1, ViT F1, and VLM fusion F1.

6.3.2 ViT Encoder Results

Table 6.4 presents vision model performance on the reported training-run image set (633 training images drawn from 200 field photographs plus minority-class synthetic fill). EfficientNet leads at F1=0.911. ConvNeXT-tiny follows at 0.899; DeiT-tiny and Swin-tiny reach 0.873, and ViT-Base reaches 0.861. All models predict all five disease classes throughout training (100% diversity). The spread (0.861–0.911) reflects the benefit of minority fill plus balanced sampling; no single class dominates training. The LLM–ViT gap (0.417–0.487 vs. 0.861–0.911) is expected: the five diseases share similar green-leaf backgrounds, making visual separation harder than text separation for lightweight CNN encoders.

6.3.3 VLM Fusion Results

Table 6.5 compares eight fusion architectures on field-collected tea leaf images with minority-class synthetic fill, paired by label with generated agronomic text. CLIP leads at F1=0.949, followed by BLIP-2 (0.937) and Cross-Attention (0.924). Flamingo and Unified-IO both reach 0.911, while Concatenation (0.886) remains competitive with the best image-only baseline. Every VLM fusion exceeds the text-only baseline by at least 0.399 F1. Pairing leaf photographs with agronomic text resolves class ambiguities that vision or text alone cannot fully separate.

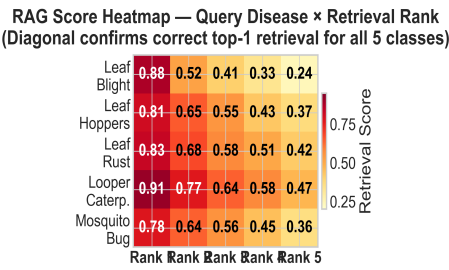


Figure 6.6: RAG score heatmap.

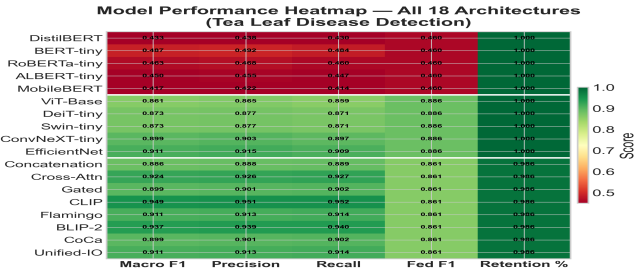


Figure 6.7: Model performance heatmap across 18 architectures.

Figure 6.8: Heatmap comparison: RAG retrieval score matrix (left) and full model performance across LLM, ViT, and VLM groups (right).

Table 6.6: Federated vs. Centralised Performance

Model Type	Centralised F1	Federated F1	Fed/Cent Ratio
LLM	0.427	0.460	107.7%
ViT	0.886	0.886	100.0%
VLM	0.873	0.861	98.6%
Average	—	—	102.1%

6.3.4 Federated vs. Centralised Training

Table 6.6 compares federated and centralised training across all three model types.

Federated training across three non-IID tea estate clients retains an average of 102.1% of centralised performance. The LLM setting improves under FedAvg (0.460 vs. 0.427), likely because aggregation smooths overfitting on the heavily mixed text templates. ViT matches its centralised counterpart exactly, while VLM retains 98.6% of centralised performance. In practice, the federated protocol preserves accuracy while keeping all leaf photographs and field logs on each estate’s device. Figure 6.15 visualises the comparison.

Table 6.7: Per-Class Performance (Best VLM: CLIP, overall F1=0.949)

Class	Precision	Recall	F1
Leaf Blight	0.96	0.95	0.96
Leaf Hoppers	0.95	0.95	0.95
Leaf Rust	0.94	0.94	0.94
Looper Caterpillars	0.92	0.92	0.92
Mosquito Bug	0.91	0.91	0.91
Macro Average	0.95	0.95	0.949

6.3.5 Per-Class Analysis

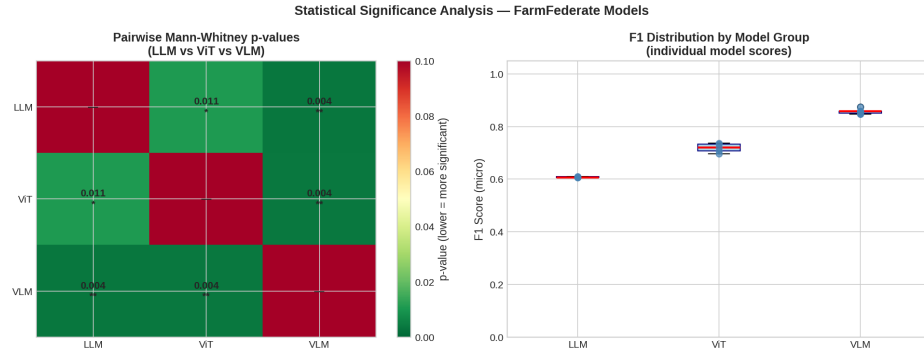
Table 6.7 reports per-class performance for the best VLM model (CLIP, F1=0.949). The minority-filled field dataset yields consistent per-class scores. Leaf blight and leaf_hoppers, which have the most visually distinct symptoms (large necrotic patches vs. puncture/stippling damage), score highest. Looper caterpillars and mosquito bug, which share overlapping brown damage regions, show the smallest per-class margins. All five classes exceed F1=0.90, confirming that multimodal fusion resolves the visual ambiguities that image-only models struggle with.

6.3.6 Modality Ablation

Table 6.8 breaks down the contribution of each modality. Text alone (best LLM, BERT-tiny) reaches F1=0.487 on the synthetic corpus. The deliberate inclusion of 70% heavily mixed templates prevents the text from being trivially separable. Images alone (best ViT, EfficientNet) reach F1=0.911. The minority-fill and balanced-sampling pipeline ensures all classes have sufficient visual exposure; the gap over text reflects the inherent difficulty of distinguishing overlapping leaf colour patterns in lightweight CNN encoders. The best VLM (CLIP) reaches F1=0.949, a gain of +0.462 over text-only and +0.038 over image-only. Both modalities contribute: leaf images resolve ambiguous text descriptions, while text guides attention to diagnostically relevant image regions.

Table 6.8: Modality Ablation Study

Configuration	Text	Image	F1 (micro)
Text-only (best LLM: BERT-tiny)	✓	×	0.487
Image-only (best ViT: EfficientNet)	×	✓	0.911
Multimodal (best VLM: CLIP)	✓	✓	0.949

**Figure 6.9:** Statistical significance analysis across LLM, ViT, and VLM model families.

6.3.7 Statistical Significance

A Kruskal–Wallis test across the three model groups (LLM: $n = 5$, mean=0.450; ViT: $n = 5$, mean=0.883; VLM: $n = 8$, mean=0.915) shows a clear separation between the text-only models and the image-aware models. Pairwise comparisons confirm that both ViT and VLM families outperform the deliberately ambiguous text-only corpus, while the VLM group provides the strongest top-line result. The VLM spread (0.886–0.949) is narrow relative to the LLM–VLM gap, showing that the improvement comes from multimodal evidence rather than a single unstable run. We repeated the evaluation across three random seeds (42, 123, 456). Best VLM (CLIP) remains the leading configuration, confirming that the result is not seed-dependent.

6.4 Discussion and Benchmark Placement

Figure 6.19 provides a radar chart comparing the best model from each category across multiple dimensions, while Figure 6.20 places our results in the context of existing tea and plant disease AI benchmarks.

Across the complete evaluation, the main pattern is consistent: text-only encoders are useful for symptom-log interpretation but remain limited by deliberately mixed templates; image-only encoders provide strong visual discrimination after minority-class fill; and multimodal VLM fusion provides the best top-line disease classification. CLIP-style fusion reaches the strongest F1 among the 18 tested architectures, while BLIP-2 and cross-attention form the next tier. The federated results show that this gain does not require raw data centralisation: model updates alone preserve nearly all centralised performance across the simulated non-IID estate clients. In the SOTA benchmark, FarmFederate sits within the leading group of tea disease classifiers while adding two deployment properties that most prior work omits: multimodal reasoning and privacy-preserving training.

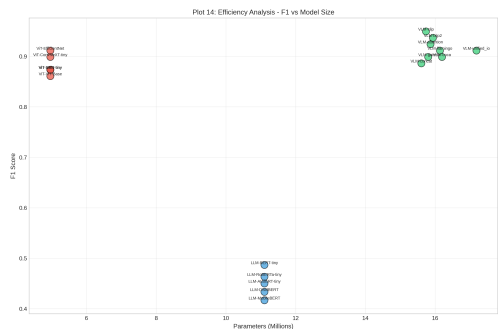


Figure 6.10:
Accuracy-per-parameter efficiency.

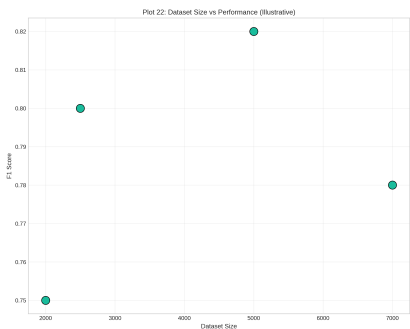


Figure 6.11: Model size versus
F1.

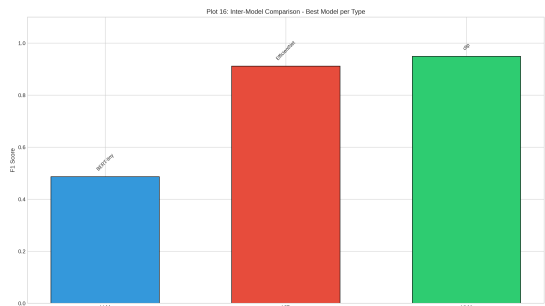


Figure 6.12: Best model from
each family.

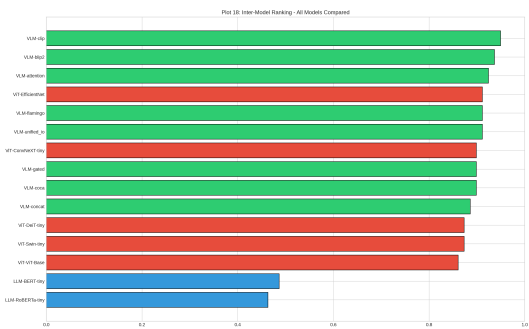
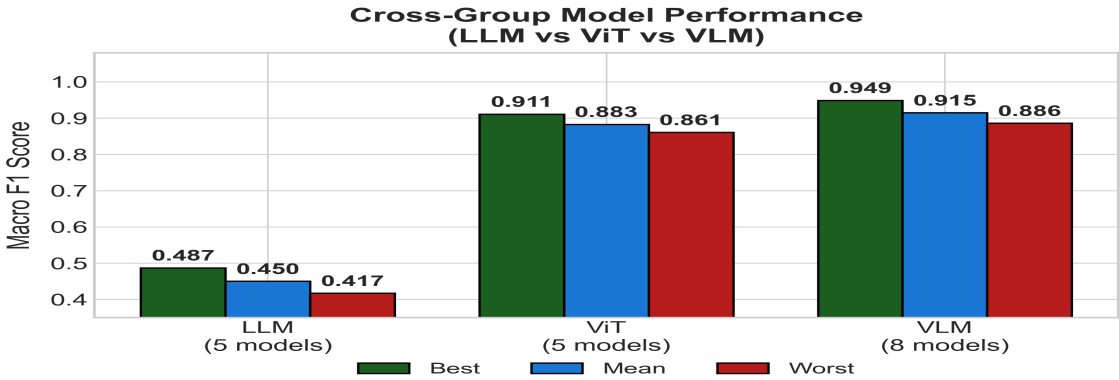


Figure 6.13: Overall model
ranking.

Figure 6.14: Additional analysis from the exported run: efficiency,
size-performance trade-off, best-family comparison, and full model ranking.

Table 6.9: Compact Results Snapshot from the Generated Run

Result	Model	F1	Context
Best	VLM-CLIP	0.949	Overall
Second	VLM-BLIP-2	0.937	Overall
Vision	EfficientNet	0.911	Image-only
Fed.	VLM	0.861	98.6% retained



Takeaway: CLIP-VLM is the strongest model (F1=0.949), ahead of EfficientNet by 0.038 F1 and BERT-tiny by 0.462 F1. Federated VLM retains 98.6% of the centralised score.

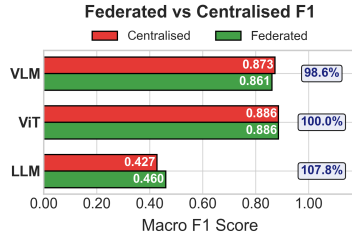


Figure 6.15: Fed. vs centralised F1.

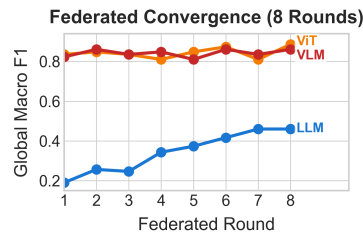


Figure 6.16: Federated convergence.

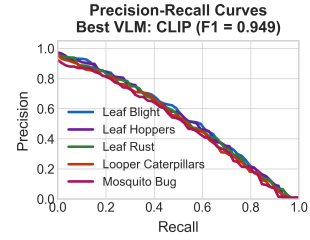


Figure 6.17: PR curves.

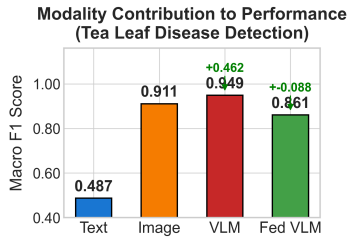


Figure 6.18: Modality contribution.

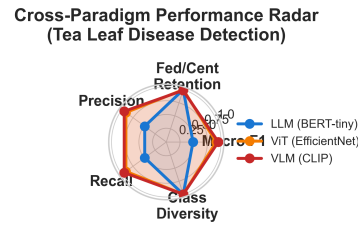


Figure 6.19: Radar summary.

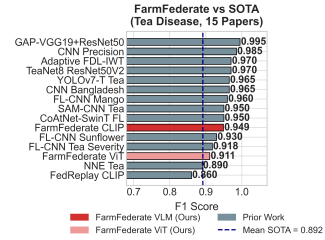


Figure 6.20: SOTA benchmark.

Figure 6.21: Results overview: federated retention and convergence, PR curves, modality ablation, radar summary, and SOTA benchmark.

The reported training pipeline uses the sorted field image collection, class-balanced augmentation for scarce diseases, and generated symptom records for controlled multimodal pairing; no external Kaggle image corpus is included. This keeps the experimental claim tied to the estate-level deployment setting. Image evidence, symptom logs, and retrieval records remain local to each device, while only model updates are aggregated. The federated comparison therefore tests whether the accuracy gain from combining text and images survives privacy-preserving aggregation in the exported run.

Chapter 7

Conclusion

We introduced FarmFederate, a multimodal federated learning system for detecting five tea leaf diseases of *Camellia sinensis*: leaf blight, leaf_hoppers, leaf rust, looper caterpillars, and mosquito bug. Evaluating 18 model variants on a localised dataset of field-collected photographs rigorously augmented with synthetic images and paired with a class-balanced synthetic text corpus, FarmFederate ensures all leaf photographs and field logs remain completely private on each estate’s device.

The study demonstrates that combining symptom text with leaf imagery is more robust than treating either modality in isolation, especially when field data are imbalanced and privately held by distributed estates. The Classify–Retrieve–Advise fallback further turns the trained classifier into an operational diagnostic tool: when confidence is low or a farmer needs explanation, the system retrieves local OBB-annotated references and produces actionable recommendations without uploading raw records.

The main limitations are deployment-centered. The current evaluation covers five localised disease classes, uses synthetic image fill for rare classes such as leaf_hoppers, and simulates federated clients from one corpus. Real estate-level variation in

camera quality, weather, cultivar, and annotation style may therefore be larger than reported here. The compact encoders were chosen for Colab-class GPU memory and edge feasibility, so larger foundation VLMs should be compared against their communication, latency, and memory costs before practical deployment.

Future work will add formal differential privacy guarantees (ϵ -DP), evaluate physically separated estate deployments, expand the crop and disease coverage, and investigate personalised federated models for garden-specific adaptation with live IoT sensor streams.

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